Valerie Salinas-vjz367
CS 4953
4/3/26

Detailed Annotated Bibliography and Classification:

Index:

Valerie Salinas-vjz367
CS 4953
4/3/26

Annotated Bibliography

1. Background/Foundational Papers

Primary Papers:

[1] Luong, M.-T., Pham, H., & Manning, C. D., 2015. Effective Approaches to Attention-based Neural Machine Translation. arXiv:1508.04025. https://arxiv.org/abs/1508.04025

This paper explores attention mechanisms in sequence to sequence models, introducing global and local attention variants for neural machine translation. They demonstrate that selective attention can improve translation quality and training efficiency compared to uniform attention. It is highly relevant as one of the first refinements of the original attention mechanism, providing insight into how attention weighting schemes can be structured. This helps motivate sparse attention patterns later developed in Transformers.

[2] Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. 2016. *Neural Machine Translation by Jointly Learning to Align and Translate*. arXiv:1409.0473. https://arxiv.org/abs/1409.0473

This paper introduces the concept of soft attention in sequence tosequence models for neural machine translation. At each decoding step, the model computes a weighted sum of all source hidden states, producing a context vector that informs the decoder which parts of the input to focus on. This allows the model to learn both alignment and translation jointly, without requiring explicit alignment labels. The work is foundational, laying the groundwork for all modern attention mechanisms. It also provides the conceptual basis for later developments in sparse and structured attention in Transformers.

[3] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł, & Polosukhin, I., 2017. Attention Is All You Need. https://arxiv.org/abs/1706.03762

Introduces the Transformer model that removes recurrence and uses only self attention mechanism to compute representations of its input and output. The paper breaks down the Transformers model that replaces the common encoder decoder layers with multi-headed self attention. Unlike other neural networks like RNNs and CNNs that work the data sequentially word by word Transformers are able to use parallelization that process all the words at once. They use Scaled dot-product attention that computes relationships between words using queries, keys and values. They claim that the transformer will produce faster results and reduce training time making them better than other machine learning models.

Valerie Salinas-vjz367
CS 4953
4/3/26

This paper is highly effective as it introduces the Transformer architecture and demonstrates strong performance improvements. However, its main limitation is the quadratic $O(n^2)$attention cost, which makes it inefficient for long sequences. This limitation directly motivates Sparse Transformers, which aim to reduce this computational cost.

[4] Richard E. Turner. 2026. *An Introduction to Transformers*. arXiv:2304.10557. https://arxiv.org/abs/2304.10557

This paper does an introduction break down of Transformers as a neural network architecture that is used to process sequences or sets of data. Transformers take input tokens and outputs new representations of those tokens which can be used for translation, classification or prediction. The Transformer block structure is broken down into the two main parts: Self attention (mixes info between tokens), feed-forward network (feed-forward layers refine each token's features). They explain self attention uses queries, keys, and values to look at each token and to compute how important each token is. They explain and show why transformers are so powerful because they work for so many types of data (text, images, audio), don't require task specific architectures and can process sequences in parallel. This paper is highly educational and effective for understanding the inner workings of Transformers. Its main limitation is that it does not propose novel methods, but it clearly illustrates why sparse and structured attention methods are necessary to address the computational and memory challenges of long sequences.

<u>Secondary Papers:</u>

[5] Gehring, J., Auli, M., Grangier, D., Yarats, D., & Dauphin, Y. N., 2017. Convolutional Sequence to Sequence Learning. arXiv:1705.03122. https://arxiv.org/abs/1705.03122

This paper proposes using convolutional neural networks (CNNs) with attention for sequence modeling, offering faster parallelization compared to RNNs while maintaining competitive performance. The attention mechanism is applied on top of the CNN layers to capture long range dependencies. It is important as it bridges early CNN based sequence models and attention based Transformers, showing efficient long range dependency modeling.

2. Fixed Structured Sparse Patterns

2.1 Sliding Window/Local Sparse Attention

<u>Primary Papers:</u>

Valerie Salinas-vjz367
CS 4953
4/3/26

[6] Beltagy, I., Peters, M. E., & Cohan, A., 2020. Longformer: The Long-Document Transformer. arXiv:2004.05150. https://arxiv.org/abs/2004.05150

This paper introduces the Longformer, a Transformer that uses a sparse attention pattern combining sliding-window local attention with task specific global tokens. By limiting each token to attend only to nearby tokens and a few global tokens, the model reduces the quadratic attention cost to linear in sequence length. Longformer is designed for processing long documents efficiently, enabling Transformers to handle thousands of tokens while maintaining performance. This paper is highly relevant as a state of the art sparse attention model that informs modern long context Transformer design. It demonstrates the practical utility of sliding-window sparse attention for efficient computation on long sequences.

[7] Zichuan Fu, Wentao Song, Yejing Wang, Xian Wu, Yefeng Zheng, Yingying Zhang, Derong Xu, Xuetao Wei, Tong Xu, and Xiangyu Zhao. 2025. *Sliding Window Attention Training for Efficient Large Language Models*. arXiv:2502.18845. https://arxiv.org/abs/2502.18845

The paper addresses the issue with Transformers computation growing quadratically (O(n^2)) with sequence length making them slow, costly and hard to use for long documents. They introduce a new method called sliding window attention training (SWAT), this method uses sliding window attention during training meaning instead of looking at every token each token only looks at nearby tokens. Previously methods used sliding windows only during inference and SWAT applies sliding windowing during training too. They do identify an issue and call it the Attention sink problem which softmax gives high weight to some tokens while others ignored which creates an imbalance in attention. They test SWAT and compare to standard Transformers and find that it has a similar full attention with 50% less memory and 30% faster training. This work is relevant as it demonstrates how local sparse attention can improve efficiency during both training and inference, providing a practical approach for long sequence Transformers.

Secondary Papers:

[8] Ali Hassani, Steven Walton, Jiachen Li, Shen Li, and Humphrey Shi. 2023. Neighborhood Attention Transformer. arXiv:2204.07143. https://arxiv.org/abs/2204.07143

In this paper they present the Neighborhood Attention Transformer (NAT) which is a sliding window attention mechanism for vision Transformers that focuses on each pixel to its nearest neighbors instead of all pixels globally. They state that it is faster and can be more memory efficient than standard self attention by reducing the computational complexity from quadratic to linear. The sliding window pattern model can see more areas of an image as layers stack and if a image shifts the features shift as well. They compare NAT to tother sliding window models and show that it performs better, faster and with their NATTEN library easier to parallelize. This paper is relevant as an example of local sparse attention applied to vision tasks, illustrating the benefits of sliding window patterns beyond text based Transformers.

[9] Zhenyu Bai, Pranav Dangi, Huize Li, and Tulika Mitra. 2024. *SWAT: Scalable and Efficient Window Attention-based Transformers Acceleration on FPGAs*. arXiv:2405.17025. https://arxiv.org/abs/2405.17025

In this paper we learn about SWAT, a method and hardware design to accelerate Transformer models with window attention on FPGAs or Field Programmable Gate Arrays. SWAT organizes the data to exploit the local attention pattern which reduces the cost of attention and feed-forward layers compared to classic Transformers. SWAT is also designed to make them run faster and more efficiently on special hardware. SWAT provides a 2x-10x faster interference and lower energy consumption. With SWAT transformers with window attention are more practical. This paper is relevant as it demonstrates the efficiency gains possible when executing local sparse attention on hardware, complementing algorithmic approaches for long sequence Transformers.

2.2 Block/Factorized Sparse Attention Patterns

Primary Papers:

[10] Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. 2019. Generating Long Sequences with Sparse Transformers. arXiv:1904.10509. https://arxiv.org/abs/1904.10509

This paper is focused on ways to change the Transformers that will help generate longer sequences that is not as slow or costly. Because their cost grows quadratically (n^2) with sequence length doubling the sequence is more expensive. They introduce Sparse Attention Transformers, which separate the full attention into shorter parts and are able to handle longer sequences than standard Transformers. To do this they used sparse factorizations of the attention matrix which scale down the cost to O(n[Equation]n). In addition, they also include updates to the deep learning architecture with adding more layers, recomputing attention only that is needed without saving everything, and use sparse attention kernels. They then show how their work and results of the architecture with model images that use the self attention that model long length sequences. This paper is relevant to Sparse Transformers because it established foundational methods for block/factorized sparse attention. While effective sparse attention may overlook some global dependencies, highlighting trade offs in long sequence modeling.

Secondary Papers:

Valerie Salinas-vjz367
CS 4953
4/3/26

[11] Piotr Nawrot, Robert Li, Renjie Huang, Sebastian Ruder, Kelly Marchisio, and Edoardo M. Ponti. 2026. The Sparse Frontier: Sparse Attention Trade-offs in Transformer LLMs. arXiv:2504.17768. https://arxiv.org/abs/2504.17768

This paper looks at how well different attention methods perform in large language models (LLMs) and want to know how much we can change or remove to make the models faster without affecting the accuracy. Tested 6 different sparse attention methods, multiple models, and evaluated 9 different tasks. Different methods work best with different tasks, Ex. For reading input use simpler patterns like token to global and block to block and in generating output use smarter attention to improve performance for more sparsity. They found that Sparse attention enables larger models to outperform smaller models and improve cost and ultimately is important for scaling LLMs. This paper is relevant as it provides systematic comparisons of sparse attention strategies, helping guide design choices for Sparse Transformers. Its findings emphasize that optimal sparse patterns are task dependent, showing the trade offs between efficiency and accuracy.

3. Hybrid Sparse Attention Designs

Primary Papers:

[12] Manzil Zaheer, Guru Guruganesh, Avinava Dubey, Joshua Ainslie, Chris Alberti, Santiago Ontanon, Philip Pham, Anirudh Ravula, Qifan Wang, Li Yang, and Amr Ahmed. 2021. *Big Bird: Transformers for Longer Sequences*. arXiv:2007.14062. https://arxiv.org/abs/2007.14062

In this paper they developed the BigBird Model, a sparse attention mechanism where instead of every token attending to every other token they limit attention connections which reduces complexity to O(n). BigBird uses three attention patterns the sliding window, global attention and random attention. This hybrid approach approximates full attention with linear $O(n)$complexity, while capturing local patterns, global context, and long range connections. Experiments show that BigBird can model sequences up to 8× longer than standard Transformers using much less memory. This paper is relevant as a core hybrid sparse attention design, blending local, global, and random attention. Its main limitation is the complexity of tuning the three attention patterns for different tasks.

[13] Fedus, W., Zoph, B., & Shazeer, N., 2021. Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity. arXiv:2101.03961. https://arxiv.org/abs/2101.03961

Switch Transformer introduces a mixture of experts (MoE) model where only a subset of experts are active per input, allowing massive scaling with minimal computation per token. This paper is relevant for hybrid sparse attention designs as it demonstrates sparse conditional computation at scale. Its focus is on expert based sparsity rather than token to token attention.

[14] Roy, A., Saffar, M., Parsani, G., & Belgrave, D., 2021. Routing Transformer: Flexible Attention via Sparse Mixtures. arXiv:2003.05997. https://arxiv.org/abs/2003.05997

Routing Transformer introduces dynamic routing based sparse attention, where tokens attend only to a subset of similar tokens. This allows This paper complements fixed pattern sparse models like Longformer and demonstrates hybrid sparse strategies that combine fixed and learned attention routing. both efficiency and flexibility, and it can adapt attention patterns based on the data. The paper is relevant for exploring data dependent sparse attention and complements Longformer's fixed sparse patterns.

[15] Joshua Ainslie, Santiago Ontanon, Chris Alberti, Vaclav Cvicek, Zachary Fisher, Philip Pham, Anirudh Ravula, Sumit Sanghai, Qifan Wang, and Li Yang. 2020. *ETC: Encoding Long and Structured Inputs in Transformers*. arXiv:2004.08483. https://arxiv.org/abs/2004.08483

We know that Transformers struggle with long inputs because of self attention computation time and memory. ETC or Extended Transformer Construction is introduced as a transformer architecture made to handle very long sequences and structured input better than the classic Transformer. ECT uses a global local attention mechanism to capture overarching context while local tokens attend only to nearby tokens to reduce computation. This lets ETC handle thousands of tokens with much less memory and time than classic transformers. Their experiments show that ETC performs better and more efficiently able to incorporate document structure like sections and paragraphs. This paper demonstrates hybrid sparse attention (global + local) for structured data. While highly effective for structured inputs, it may be less general purpose for other domains.

Secondary Papers:

[16] Dong Liu and Yanxuan Yu. 2025. π-Attention: Periodic Sparse Transformers for Efficient Long-Context Modeling. arXiv:2511.10696. https://arxiv.org/abs/2511.10696

This paper introduces π-Attention, as a new sparse method that is aiming to be better than others such as Ring Attention by handling far away tokens while maintaining efficiency. For π-Attention uses local attention (nearby tokens), periodic skips (jumps to tokens/every k step), and uses adaptive fusion gate (figures out how to combine local and long range info. Their results showed better prediction, fewer FLOPS, and that it scaled well to long sequences. Some issues were that they used a fixed skip patten that could drop performance and that the attention pattern is manually designed making it not at flexible as models that learn. Future plans involve making skip pattern more learnable, combine routing based attention, and apply multimodal models. Innovative combination of local and periodic attention, though fixed patterns limit flexibility. This paper is relevant as a modern hybrid sparse attention design improving long range efficiency in Transformers.

[17] Du, N., Shazeer, N., Xiong, C., Li, Z., Chen, M., & Le, Q. V., 2022. GLaM: Efficient Scaling of Language Models with Mixture-of-Experts. arXiv:2112.06905. https://arxiv.org/abs/2112.06905

GLaM (Generalist Language Model) introduces a mixture of experts (MoE) approach for large multi modal models, where each input token is routed to only a small subset of expert networks. By activating only a fraction of the model for each token, GLaM achieves high computational efficiency while scaling to very large model sizes and diverse tasks. The model demonstrates that sparse routing at the expert level can maintain or even improve performance compared to dense architectures, making it a practical solution for scaling multi-modal models. This approach relates to sparse Transformers because it applies sparsity at the model level rather than the attention level: just as sparse attention reduces computation by limiting token interactions, GLaM reduces cost by limiting which experts are active, showing that sparsity whether in attention or in model pathways is a key strategy for scaling large neural networks efficiently.

[18] Kaleab A. Kinfu and René Vidal. 2025. *Transformers with Joint Tokens and Local-Global Attention for Efficient Human Pose Estimation*. arXiv:2503.00232. https://arxiv.org/abs/2503.00232

In this paper the authors use Transformers to identify where body parts are in an image. They created a model called EViTPose that focuses on important body parts in an image while still maintaining a high accuracy. Another model called UniTransPose focuses on small details and the entire body to understand certain body poses. They combined multiple datasets together so that models would learn better and faster. The results showed that the models were accurate and efficient. Effective application in vision tasks, though domain specific. Shows how attention (often sparse/local) works in vision.

4. Adaptable/Learnable Sparsity:

<u>Primary Papers:</u>

[19] Sainbayar Sukhbaatar, Edouard Grave, Piotr Bojanowski, and Armand Joulin. 2019. Adaptive Attention Span in Transformers. arXiv:1905.07799. https://arxiv.org/abs/1905.07799

Valerie Salinas-vjz367
CS 4953
4/3/26

Transforms use a fixed attention span which could be costly. This paper proposes using Adaptive Attention Span, a mechanism that allows each attention head to learn its own context length. Instead of using a pre-defined fixed window manually, the model learns how much context it actually needs. This can reduce computation, speed up training, and improve performance. The learned spans reveal that early layers focus locally while later layers capture long range context. Adaptive attention exemplifies learnable sparsity, giving the model control over which tokens to prioritize. Its main trade off is added complexity in training.

<u>Secondary Papers:</u>

[20] Hemanth Saratchandran and Simon Lucey. 2025. *Enhancing Transformers Through Conditioned Embedded Tokens*. arXiv:2505.12789 https://arxiv.org/abs/2505.12789

There is a problem within Transformers called "ill conditioned attention" that makes training slower and harder. They propose a method called conditioned embedded tokens which changes the input embeddings to make the attention easier to train. By using this method it showed that it stabilizes training and improves performance on image classification, object detection, segmentation and natural language processing. Adjusting the embeddings it can make transformers faster and more reliable without changing the overall architecture. Using this method overall shows it can make transformer train more efficiently. Improves training stability, though adds preprocessing complexity. Helps stabilize attention learning, including sparse attention.

[21] Renpu Liu, Ruida Zhou, Cong Shen, and Jing Yang. 2025. On the Learn-to-Optimize Capabilities of Transformers in In-Context Sparse Recovery. arXiv:2410.13981. https://arxiv.org/abs/2410.13981

In this paper they aimed to understand the Transformers in context learning or (ICL) that performs well without the need for further parameter updating. The belief is that Transformers perform pre-conditioned gradient descent which allows them to learn a pre-conditioner during pre-training and then use this in the ICL to expedite and optimize the process. They focus on in context sparse recovery, where the goal is to reconstruct signals with only a few nonzero elements from limited data. They look at how Transformers implement the L20 algorithm for in context sparse recovery and show their results. The paper provides an interesting theoretical view of Transformers as optimization algorithms, though its focus is narrow (sparse recovery) and less directly applicable to general architectures. While not directly about sparse attention, it relates to how Transformers internally optimize sparse representations, which complements sparse modeling ideas.

5. Memory-Augmented or Token Augmentation Methods

<u>Primary Papers:</u>

Valerie Salinas-vjz367
CS 4953
4/3/26

[22] Sainbayar Sukhbaatar, Edouard Grave, Guillaume Lample, Herve Jegou, and Armand Joulin. 2019. *Augmenting Self-attention with Persistent Memory*. arXiv:1907.01470. https://arxiv.org/abs/1907.01470

Since Transformers rely entirely on self attention over input tokens, they may forget important information because there is no permanent memory. This paper introduces using persistent memory tokens to the Transformer, so instead of only having the input tokens the model also has a set of learnable memory vectors that will act like global storage. This will give the Transformer a memory set to be able to refer to, and it learns what should be stored during training. These memory vectors can help the model remember things giving it a longer range. This approach enhances the model's ability to handle long range dependencies, though it introduces extra memory cost. Unlike sparse attention methods that reduce computation by limiting connections, persistent memory extends the model's capacity to retain information over long sequences.

[23] Mikhail S. Burtsev, Yuri Kuratov, Anton Peganov, and Grigory V. Sapunov. 2021. *Memory Transformer*. arXiv:2006.11527. https://arxiv.org/abs/2006.11527

This paper discusses how Transformers struggle with retaining information for long sequences because self attention only uses information from the current input and doesn't have a structured memory. They propose a Memory Transformer, a modified Transformer that maintains an external memory space which grows over time. They created three extensions MemTransformer, MemCtrl, MemBottleneck and tested on multiple versions. They found that adding memory improves performance, but MemBottleneck performs worse. Ultimately adding memory can improve the Transformers' ability to process global context and complex tasks. Strong approach for long context, but introduces architectural complexity. Competes with Sparse Transformers for efficient long sequence modeling.

Secondary Papers:

[24] Ignacio Sastre and Aiala Rosá. 2025. *Memory Tokens: Large Language Models Can Generate Reversible Sentence Embeddings*. In *Proceedings of the First Workshop on Large Language Model Memorization (L2M2)*. Association for Computational Linguistics, 2025. https://arxiv.org/pdf/2506.15001v1

In the paper Large Language models (LLMs) can generate special tokens called memory tokens. These tokens function like reversible sentence embeddings, they can store a sentence in a way and later reconstruct the sentence to its original form. This is important because most embeddings cannot be reversed. Suggesting that this is a new way to store and retrieve text in models. They show in their experiments that this method can help improve LLMs memory and retrieve text accurately. Memory tokens illustrate a memory based alternative to sparsity, supporting efficient long context modeling without modifying the attention mechanism, though the method remains experimental.

Valerie Salinas-vjz367
CS 4953
4/3/26

6. Complementary or Supportive Structural Techniques

Primary Papers:

[25] Dao, T., Li, A., Vaswani, A., Ho, D., Pham, H., Le, Q., & Le, H., 2022. FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness. arXiv:2205.14135. https://arxiv.org/abs/2205.14135

FlashAttention is an optimized attention kernel that reduces memory usage and increases speed by reordering computation and IO patterns. It allows exact attention computation for very long sequences without approximate methods. This paper is relevant as it addresses hardware-level efficiency improvements for sparse and dense attention models, crucial for scaling Transformers.

Secondary Papers:

[26] Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, L., Wang, L., Chen, W., & Chen, L., 2021. LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685. https://arxiv.org/abs/2106.09685

LoRA proposes injecting low rank adapters into pre-trained models, allowing task specific fine tuning with few additional parameters. It enables scaling to large models efficiently without retraining all parameters. This is highly relevant as a parameter efficient adaptation method, complementing sparse attention and memory augmented approaches for large-scale Transformers.

[27] Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., de Laroussilhe, Q., Gesmundo, A., Attariyan, M., & Gelly, S., 2019. Parameter-Efficient Transfer Learning for NLP. arXiv:1902.00751. https://arxiv.org/abs/1902.00751

This paper introduces adapter layers, small trainable modules added to pre-trained models. Only the adapters are updated during fine tuning, allowing efficient domain adaptation with minimal additional parameters. The work is relevant for parameter efficient scaling, showing that task specific adaptation can be decoupled from the base model, which complements LoRA.

[28] Mary Phuong and Marcus Hutter. 2022. Formal Algorithms for Transformers. arXiv:2207.09238. https://arxiv.org/abs/2207.09238

This paper focuses on the mathematical computations that make Transformers perform well they present algorithms with explanation and pseudocode. They address the fact that there is not much publication of source and pseudocode within the deep learning community which is the motivation for their paper. They break down tokenization and multiple algorithms used in Transformers with explanation and pseudocode. Then they describe different Transformer architectures and explain how different attention mechanisms work. The paper's purpose is to provide a solid understanding of Transformers and to use the pseudocode as templates for future contribution. Valuable for understanding implementation details, but not introducing new architectures. Helps explain how sparse attention mechanisms can be implemented.

[29] David R. So, Chen Liang, and Quoc V. Le. 2019. The Evolved Transformer. arXiv:1901.11117. https://arxiv.org/abs/1901.11117

In this paper they wanted to know if a machine would produce a better version of the Transformer by searching through many possible architectures automatically instead of manually designing one. The goal was to take Neural Architecture Search (NAS), which was outperforming human designed models, and apply NAS to search for a better Transformer architecture. They introduced Progressive Dynamic Hurdles(PDH), a method that will give more compute to the better models and stop training the weaker models early. They were able to find that it was successful and found a Transformer variation that performed better calling it the Evolved Transformer. Shows NAS can improve architectures, though computationally expensive to search. Could be extended to discover sparse attention patterns automatically.

[30] Subhabrata Dutta, Tanya Gautam, Soumen Chakrabarti, and Tanmoy Chakraborty. 2021. *Redesigning the Transformer Architecture with Insights from Multi-particle Dynamical Systems*. arXiv:2109.15142. https://arxiv.org/abs/2109.15142

Transformers work well however they are very large with too many parameters and are slow because attention compares every token with every other token which makes them costly to train and use. The goal for this paper is they want to simplify its attention and feed forward layers to make it smaller and faster. They introduce a new way to thinking about Transformers and treat it like a multi-particle dynamical system and connect this to the ordinary differential equations (ODEs). They call it the TransEvolve method and instead of tokens attending to each other the tokens move and influence each other like particles in physics. They test this and find that tokens behave like interacting particles reducing computation, parameters and more efficient approximation of attention. Creative theoretical framing, though less widely adopted in practice. Another way to reduce computation without explicit sparsity.

[31] Zhijian Zhuo, Yutao Zeng, Ya Wang, Sijun Zhang, Xiaoqing Li, Jian Yang, Zhou Xun, and Jinwen Ma. 2025. *HybridNorm: Towards Stable and Efficient Transformer Training via Hybrid Normalization*. In *The Thirty-ninth Annual Conference on*

Neural Information Processing Systems. https://openreview.net/forum?id=NligLHO7yG

The paper states the issue with Transformers and claims its due to how normalization layers are used. They discuss the two common normalization strategies: Pre-Norm being more stable, easier gradient flow but worse final performance and Post-Norm has better performance but instable training which could fail. They propose mixing the two inside a Transformer block and call it HybridNorm. HybridNorm uses QKV Normalization which normalize query, key, value separately and Post-Norm after feed forward network. They found that HybridNorm showed better results and that using multiple normalization styles made for better accuracy and lower loss. The method improves gradient flow and reduces training instability, making it easier to train sparse Transformers reliably across deep architectures.

[32] Brian Lester, Rami Al-Rfou, and Noah Constant. 2021. *The Power of Scale for Parameter-Efficient Prompt Tuning*. arXiv:2104.08691. https://arxiv.org/abs/2104.08691

In this paper they want to know how much task specific information can be learned by tuning only a small number of parameters. They propose prompt tuning as a parameter efficient alternative to full fine tuning. They experiment with different model sizes find that prompt tuning works better as models get larger and almost match full fine tuning and small models benefits less. Comparing to full fine tuning they see that prompt tuning requires <1% of the parameters. Ultimately it is simpler, scalable alternative to fine tuning. Prompt tuning demonstrates a scalable way to reduce computational cost, offering a complementary strategy to sparse attention for improving efficiency in large Transformer models.

[33] Bobby He and Thomas Hofmann. 2024. *Simplifying Transformer Blocks*. arXiv:2311.01906. https://arxiv.org/abs/2311.01906

The paper claims that even though Transformer blocks appear simple they can be sensitive by even the smallest changes when training which could slow it down. They want to know how much from the architecture we can remove before it disrupts performance. They introduce a new version for attention and called it Simplified Attention Sub-block (SAS), which removed some parameters and simplified attention computation. In the Parallel Block Design changed the attention from feed forward to running them in parallel which made the model simpler. They were able to show in their results that the simplified Transformer worked just as well as the standard Transformer and was faster with less resources. This work illustrates how architectural simplification can reduce computational cost, offering an alternative or complement to sparse attention mechanisms.

Valerie Salinas-vjz367
CS 4953
4/3/26

[34] Teodor-George Marchitan, Claudiu Creanga, and Liviu P. Dinu. 2024. *Transformer and Hybrid Deep Learning Based Models for Machine-Generated Text Detection*. arXiv:2405.17964. https://arxiv.org/abs/2405.17964

They begin by stating the need to detect Machine Generated Text (MGT) now that there have been advances in large language models. They want to take Transformer based architectures and smart hybrid models for detecting MGT. Combining Transformers with CNNs or LSTMs capture local sequential and global features at the same time making them better at detecting MGT. From their experiments they find that Transformers outperform standard models especially RoBERTa based models achieve the best results at finding longer sequences. While the paper emphasizes application rather than architectural innovation, it demonstrates the practical value of Transformer models in handling sequential and structural patterns efficiently, reinforcing why sparsity and attention optimizations can enhance performance on tasks.

[35] Eren Unlu. 2024. *Geotokens and Geotransformers*. arXiv:2403.15940. https://arxiv.org/abs/2403.15940

In Geotokens and Geotransformers the authors want to help transformers understand geographic locations better. Transformers use position encodings for work in a sentence but they don’t capture actual instances. They created Geotokens, which are tokens that represent locations with latitude and longitude. This is significant because it has the Transformer focus on the tokens based on distance instead of order. With this Transformers can become better when it comes to maps, GPS data or physical location. By structuring attention according to distance, Geotransformers offer an example of implicit sparsity, showing how attention patterns can be constrained and guided for efficiency in specialized contexts.

[36] Jerome Garnier-Brun, Marc Mézard, Emanuele Moscato, and Luca Saglietti. 2025. *How Transformers Learn Structured Data: Insights from Hierarchical Filtering*. arXiv:2408.15138. https://arxiv.org/abs/2408.15138

In this paper they want to understand how Transformers learn and represent structured or hierarchical data. The authors introduce the concept of hierarchical filtering, where early layers capture coarse structure and deeper layers focus on fine details. They showed that Transformers naturally refine hierarchical relationship through layers by analyzing attention weights and by experiments on synthetic and real language tasks. Which would explain why Transformers are good at nested or structed patterns like syntax. In this paper we understand what happens when Transformer modes learn structured data. Understanding this layered attention behavior informs sparse attention design, as limiting attention to relevant subsets of tokens can effectively preserve hierarchical information while reducing computation.

[37] Weixin Liang, Lili Yu, Liang Luo, Srinivasan Iyer, Ning Dong, Chunting Zhou, Gargi Ghosh, Mike Lewis, Wen-tau Yih, Luke Zettlemoyer, and Xi Victoria Lin. 2025.

Valerie Salinas-vjz367
CS 4953
4/3/26

Mixture-of-Transformers: A Sparse and Scalable Architecture for Multi-Modal Foundation Models. arXiv:2411.04996. https://arxiv.org/abs/2411.04996

Mixture of Transformers or MoT is a scalable architecture that uses a sparse mixture of Transformer experts instead of just one large model. In this a router network sends different tokens to only a few relevant experts saving computation and memory. MoT is able to handle large multi-modal datasets while improving performance and accuracy. MoT out performs models of similar size and is able to scale big tasks as well. MoT demonstrates that sparsity can be applied at the model level, not just within attention, offering a practical approach to large scale Transformer efficiency.

[38] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. 2019. *Parameter-Efficient Transfer Learning for NLP*. arXiv:1902.00751. https://arxiv.org/abs/1902.00751

In this paper we learn a new way to fine tune parameters called Parameter Efficient Transfer Learning that keeps the original model mostly fixed and only trains a small number of extra parameters for each task. This is important because pre-trained models are huge and trying to fine tune all parameters for every model is costly. We can adapt large pre-trained language models to new tasks without updating all their parameters, and instead we can insert small modules called adapter layers between the transformer layers and only train those. Adapter models perform close to full fine tuning while using fewer trainable parameters. With this transfer learning is more memory efficient and preferred when applying the same base model to multiple tasks. This approach complements sparse attention strategies by demonstrating another way to make large Transformer models more efficient without changing their core attention mechanisms.